

# Grade 3 Maths: Angles

Measuring with a Protractor

Subject: Mathematics

Grade: 3

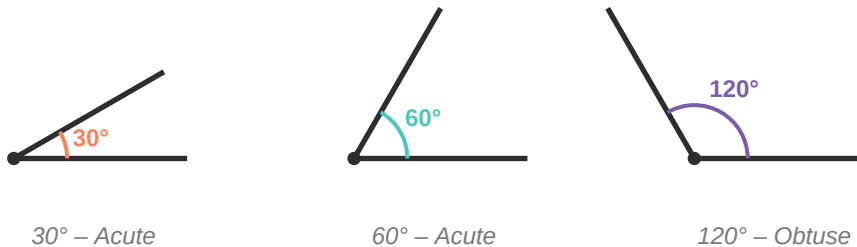
Topic: Angles & Protractors

## ■ Learning Objectives

- ✓ Define what an angle is and identify it in everyday objects.
- ✓ Name the parts of a protractor (curved edge, center hole, degree markings).
- ✓ Follow the 4-step method to measure angles accurately.
- ✓ Distinguish between the inner scale and the outer scale.
- ✓ Avoid common measurement mistakes.

## ■ What Is an Angle?

An **angle** is the space between two straight lines (called **rays**) that meet at a common point called the **vertex**. We measure angles in **degrees (°)**.



## ■ Meet the Protractor

## ■ Meet the Protractor

|                 |                                                                          |
|-----------------|--------------------------------------------------------------------------|
| Center Hole     | Place this exactly on the vertex of the angle.                           |
| Curved Edge     | The semicircular rim that holds the degree markings.                     |
| Degree Markings | Numbers from $0^\circ$ to $180^\circ$ on both the inner and outer scale. |
| Baseline        | The straight flat edge at the bottom; lined up with one ray.             |
| Inner Scale     | Reads left-to-right ( $0^\circ$ on the right side).                      |
| Outer Scale     | Reads right-to-left ( $0^\circ$ on the left side).                       |

### ⇒ How to Measure an Angle – 4 Steps

- 1 Place the Protractor**  
Put the center hole of the protractor exactly on the vertex — the point where the two lines of the angle meet. Make sure it doesn't slip!
- 2 Align the Baseline**  
Rotate the protractor so that the straight edge (baseline) lines up perfectly with one ray of the angle at the  $0^\circ$  line.
- 3 Read the Degree Mark**  
Look where the other ray crosses the curved scale. Read the number — that is your angle! Make sure you use the correct scale (inner or outer).
- 4 Write It Down**  
Record the measurement and always include the degree symbol  $^\circ$ . For example:  $45^\circ$ ,  $90^\circ$ ,  $120^\circ$ .

■ **Inner vs Outer Scale Tip:** If the angle looks **smaller than  $90^\circ$** , use the **smaller number**. If the angle looks **larger than  $90^\circ$** , use the **larger number**.

## ■ Common Mistakes to Avoid

|                                  |                                                                      |
|----------------------------------|----------------------------------------------------------------------|
| ✗ <b>Wrong center placement</b>  | The center hole must sit exactly on the vertex, not beside it.       |
| ✗ <b>Baseline not at zero</b>    | Always line up one ray with $0^\circ$ before reading the scale.      |
| ✗ <b>Reading the wrong scale</b> | Protractors have two sets of numbers. Choose the right one!          |
| ✗ <b>Protractor slipping</b>     | Hold the protractor firmly or use a pencil to mark the vertex first. |

|                                  |                                                                      |
|----------------------------------|----------------------------------------------------------------------|
| ■ <b>Baseline not at zero</b>    | Always line up one ray with $0^\circ$ before reading the scale.      |
| ■ <b>Reading the wrong scale</b> | Protractors have two sets of numbers. Choose the right one!          |
| ■ <b>Protractor slipping</b>     | Hold the protractor firmly or use a pencil to mark the vertex first. |

## ■ Tips for Accurate Measuring

- ■ Always double-check the vertex placement before reading.
- ■ Use a sharp pencil to draw clear, thin lines.
- ■ Keep your protractor clean and flat against the paper.
- ■ Practice with many different angles to build confidence.
- ■ Estimate first — does your answer look about right?

## ■ Fun Facts About Angles!

Angles are all around us every day! The hands of a **clock** form angles, a **pizza slice** is a triangle with angles, **kite tails** make angles in the sky, and even butterfly wings have angles. Can you spot five angles in your classroom right now?

## ■ Practice Questions

1. What is an angle?

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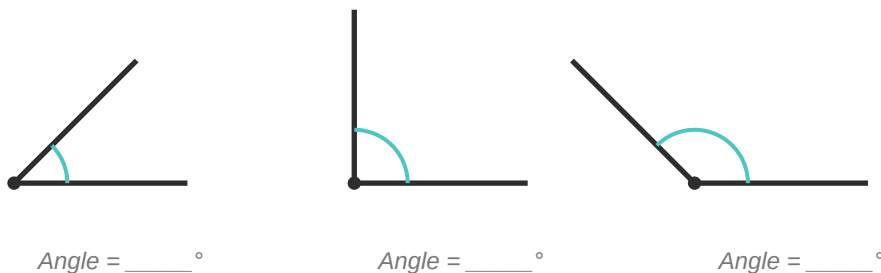
2. Name THREE parts of a protractor.

a) \_\_\_\_\_ b) \_\_\_\_\_ c) \_\_\_\_\_

3. What is the FIRST step when measuring an angle with a protractor?

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4. Look at the diagram below. Write the correct measurement for each angle.



5. A student measured an angle and wrote  $130^\circ$ , but the angle looks smaller than  $90^\circ$ . What mistake did they likely make?

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6. True or False? You should always write the degree symbol  $^\circ$  after your measurement.

Answer: \_\_\_\_\_

7. Circle the correct answer: An angle that measures exactly  $90^\circ$  is called a a) Straight angle b) Right angle c) Acute angle d) Obtuse angle
8. Draw an angle of  $45^\circ$  in the space below.

### ■ Answer Key (Teacher Copy)

|     |                                                                                                       |
|-----|-------------------------------------------------------------------------------------------------------|
| Q1. | An angle is the space between two lines that meet at a point (the vertex).                            |
| Q2. | Any three of: Center hole, Curved edge, Degree markings, Baseline, Inner scale, Outer scale.          |
| Q3. | Place the center hole of the protractor exactly on the vertex of the angle.                           |
| Q4. | a) $45^\circ$ b) $90^\circ$ c) $135^\circ$                                                            |
| Q5. | The student read the wrong scale on the protractor (used the outer scale instead of the inner scale). |
| Q6. | True                                                                                                  |
| Q7. | b) Right angle                                                                                        |

|            |                                                                                           |
|------------|-------------------------------------------------------------------------------------------|
| <b>Q8.</b> | Check that the student's drawn angle measures $45^\circ$ when verified with a protractor. |
|------------|-------------------------------------------------------------------------------------------|

### ■ Summary: 4 Steps to Measure an Angle

- 1■■ Place the center hole on the vertex.
- 2■■ Align one ray with the  $0^\circ$  baseline.
- 3■■ Read the degree mark where the other ray crosses the scale.
- 4■■ Write down the measurement with the  $^\circ$  symbol.